Khulna University of Engineering & Technology

Department of Electronics and Communication Engineering

Khulna, Bangladesh

PROJECT PROPOSAL

FASTEST FINGER QUIZ SYSTEM USING DIGITAL LOGIC ICS

Course No: ECE 2104

Course Title: Digital Electronics and Logic Circuits Laboratory

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Project Title:

Design and Implementation of a Fastest Finger Quiz System Using Digital Logic ICs

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Introduction

In quiz competitions, identifying the fastest participant to respond is essential for fairness and efficiency. Conventional quiz systems often rely on microcontrollers or software-based solutions. However, understanding hardware logic implementation remains an important part of digital electronics education.

This project proposes the design and development of a **fully hardware-based Fastest Finger Quiz System** implemented using standard digital logic integrated circuits (ICs). The system determines the first responding player within a limited time period and prevents multiple responses using hardware locking mechanisms.

The project demonstrates practical applications of sequential logic, combinational logic, timing control, and input prioritization without using any microcontroller or programming.

Objectives of the Project

The main objectives of this project are:

- To design a hardware-based quiz buzzer system.
- To implement a 0–9 timing counter using digital ICs.
- To detect and display the fastest responding player.
- To allow only one valid response per round.
- To automatically disable inputs after a valid response or timeout.
- To indicate response status using visual indicators (LEDs and displays).
- To gain practical knowledge of digital electronics circuit design.

Problem Statement

During quiz competitions, multiple participants may attempt to answer simultaneously. Manual judgement may cause unfair decisions or confusion. Therefore, an automated system is required that:

- Accurately detects the first button press.
- Prevents further inputs once a response is registered.
- Provides a limited response time.
- Clearly indicates whether a valid response occurred or time expired.

Proposed System Description

The proposed system consists of two major functional sections:

i) Timing and Counting Unit

- A reset button starts the quiz round.
- A counter begins counting from **0 to 9**.
- The counting duration represents the allowed answering time.
- The count value is displayed using a **7-segment display**.
- The counter automatically stops when:
    - a player presses a button, or
    - the count reaches 9 without any response.

ii) Player Response Control Unit

- Four players are connected through individual push buttons.
- When the timer starts, all players are enabled.
- The **first player** who presses the button is registered.
- The system immediately:
    - locks all other player inputs,
    - stops the counter,
    - displays the player number (1–4) on another 7-segment display,
    - turns ON a **green LED** indicating a valid response.

iii) Timeout Operation

If no player presses any button before the counter reaches 9:

- Player inputs are automatically disabled.
- A **red LED** turns ON to indicate timeout.
- The system waits for the reset button to start a new round.

iv) Reset Operation

- A reset push button initializes the system.
- Counter returns to zero.

- Player circuits become active again.
- The next quiz round can begin.

Methodology

The project will be implemented using digital electronics design techniques:

1. Designing block diagram of the system.
2. Developing timing circuit using clock generator.
3. Implementing counter circuit.
4. Designing input priority and locking mechanism.
5. Interfacing 7-segment displays using decoder drivers.
6. Integrating LEDs for status indication.
7. Testing and debugging the complete hardware system.

Required Components

SL.No.	Name of instruments	Ratings	Quantity
1	Voltage Supply	0~5 volts	1
2	555 timer	NE555	5
3	Decade counter	74LS90	1
4	BCD to 7-segment display Decoder	74LS48	2
5	LED	Red(1), Green(4), Blue(1)	6
6	Resistors	10k, 15k, 68k, 1k	~
7	Capacitors	10uF - 1pcs 0.01uF - 1pcs	2

8	Diodes	1N4148	4
9	Push button		5
10	7-segment display	common cathode	2
11	AND	7408	2
12	OR	7432	1
13	NOR	7402	1
14	NOT	7404	1
15	J-K flip-flop	7476	2
16	(dual input) NAND	7400	1
17	(Quad input) NAND	7413	1
18	Breadboard	16.5cm × 5.5cm 830 points total	4

System Design

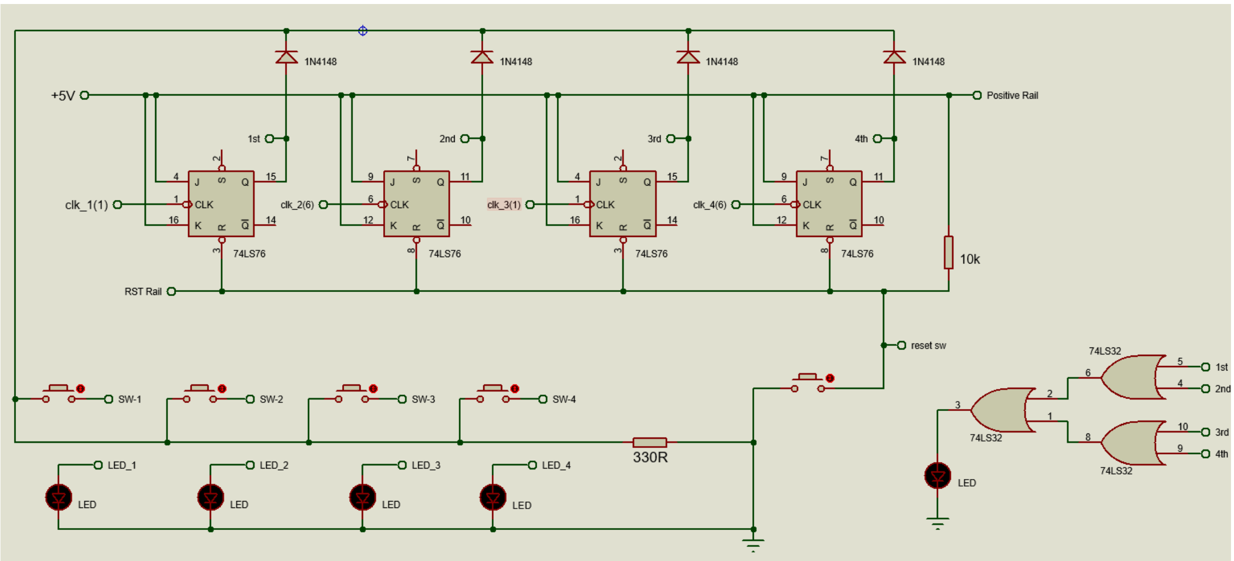

Fig : Player Response Control Circuit [with JK filp-flops].

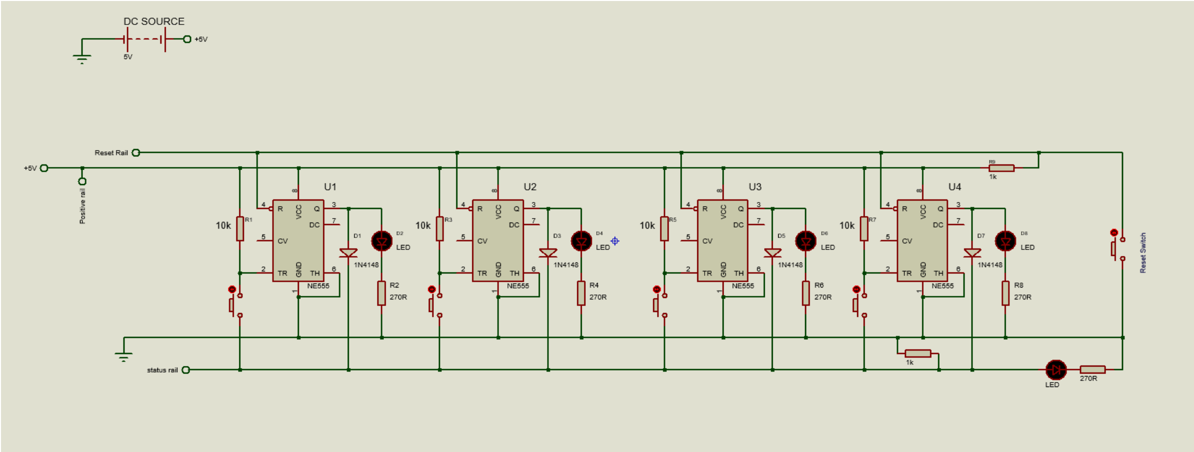

Fig : Player Response Control Circuit [with 555 timer].

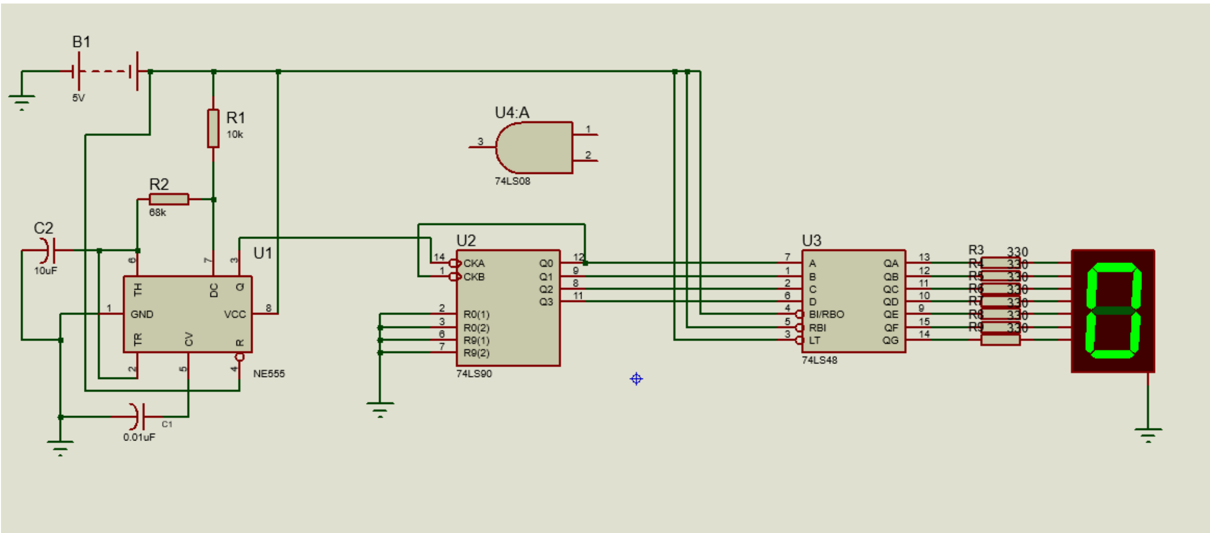

Fig : Timing and Counter Circu

Expected Outcomes

After successful implementation, the system will:

- Accurately detect the fastest player.
- Display response time visually.
- Prevent multiple answers.
- Automatically handle timeout conditions.
- Operate completely without software or microcontroller.

Applications

- Academic quiz competitions
- Classroom demonstrations
- Electronics laboratory experiments
- Digital logic education systems
- Competitive game shows

Advantages of the Proposed System

- Fully hardware-based operation
- No programming required
- Fast and reliable response detection
- Low cost and easy maintenance
- Educational demonstration of digital logic concepts

Conclusion

The proposed Fastest Finger Quiz System provides a practical implementation of digital electronics concepts through a real-world application. By eliminating software dependence and using only hardware logic ICs, the project enhances understanding of timing circuits, counters, input prioritization, and sequential logic design. The system aims to ensure fairness, accuracy, and automation in quiz competitions while serving as an effective educational platform.